Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/01/2013
PI Name:	Chen, David J. Ph.D.		
Project Title:	Mechanisms of the Repair of HZE Induced DNA Double-Strand Breaks in Human Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	05/10/2012	**End Date:**	12/09/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Asaithamby, Aroumougame (University of Texas Southwestern Medical Center)		
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	[Editor's note 5/29/2012] Note this is a continuation of "Mechanisms of the Repair of HZE Induced DNA Double-Strand Breaks in Human Cells" (grant NNX07AU42G). During our upcoming beam times at the National Space Radiation Laboratory, Brookhaven National Lab, New York, we will test two major hypotheses: 1. Cells at different phases of cell cycle repair clustered DNA lesions induce HZE particles differently. 2. The nature of clustered DNA lesions induced by HZE particles is dependent of the physical properties of the particles. Experiment #1: Establishment of a base damage marker for live cell imaging and evaluation of kinetics of clustered		

Task Description:	DNA lesions in live cells. (i). Identify a suitable base damage marker for live cell imaging. (ii). Evaluate the kinetics of recruitment of different DNA repair factors to the sites of DNA lesions induced by HZE particles in live cells. Experiment #2: Establish approaches for cell synchronization and study induction and repair of clustered DNA lesions at different phases of cell cycle. (i). Synchronize cells at different phases of cell cycle. (ii). Identify different phases of cell cycle. (iii). Study induction and repair of clustered DNA lesions at different phases of cell cycle. (iv). Determine cell cycle-dependent recruitment and retention of DNA repair proteins at the sites of clustered DNA lesions. (v). Determine whether there is a differential repair capability of clustered DNA lesions in G1 vs. S phase of the human HT1080 cells. (vi). Monitor clustered DNA lesions repair in G2 phase of the cell cycle. Experiment #3: Quantification and imaging analysis of clustered DNA lesions to determine whether the nature of DNA lesions induced by HZE particles is dependent on the physical properties of the HZE particles. These experiments will be carried out by the Co-PI. (i). Quantify charge and LET dependent induction and repair of clustered DNA lesions. (ii). Determine the nature of clustered DNA lesions induced by different HZE particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research would lead to the development of counter-measurement to enhance astronaut's ability to sustain HZE particles and radiation exposures in space
Task Progress:	1. Homologous repair deficient hamster cells are highly sensitive to iron particles as compared with wild-type cells. 2. WRN-deficient human cells are extremely sensitive to Fe particles as compared with gamma-rays. As DSBs induced by HZE particles are complex, and WRN has shown to have multiple roles in NHEJ, HR and BER pathways, it might play one or more roles in HZE particle induced DNA repair either by coordinating within the individual DNA repair pathways or it may coordinate between them. This could be achieved either by binding directly to the damaged DNA or recruited by other DNA-repair factors. 3. Artemis-deficient human cells are significantly sensitive to iron particles as compared with wild-type cells. However, the mechanism by which Artemis participates in the processing of complex DNA lesions is not clear. Artemis might be needed to process damaged ends at DSBs prior to rejoining. This might require DNA-PK- and ATM-dependent phosphorylation of Artemis followed by its activity modification from exonuclease to endonuclease. 4. FANCA-deficient human cells are highly sensitive to Fe particles as compared with FANCA deficient cells complemented with WT FANCA, suggesting a role for FA complex in the repair of clustered DNA lesions. However, whether all the factors of the FA complex orchestrate the repair of complex DNA lesions and the method by which these factors function in this processes is totally unknown.
Bibliography Type:	Description: (Last Updated: 08/14/2018)
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "Biological consequences of unrepaired clustered DNA lesions in response to high-atomic number and energy particles in monolayer and organotypic cell cultures." Presented at the 39th Annual Meeting of the European Radiation Research Society, Vietri sul Mare, Italy, October 15-19, 2012. 39th Annual Meeting of the European Radiation Research Society, Vietri sul Mare, Italy, October 15-19, 2012. Abstract book, p. 72. , Oct-2012
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "Mechanism of HZE Particles Induced Chromosome Instability." Presented at the 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. (Oral Presentation.) 23rd Annual NASA Space Radiation Investigators' Workshop, Durham, NC, July 8-11, 2012. , Jul-2012
Abstracts for Journals and Proceedings	Asaithamby A, Chen DJ. "Unrepaired clustered DNA lesions together with defective G2/M checkpoint induce chromosome instability in human cells." Presented at the 10th International Symposium on Chromosomal Aberrations, Amalfi, Italy, October 19-21, 2012. 10th International Symposium on Chromosomal Aberrations, Amalfi, Italy, October 19-21, 2012. Abstracts. , Oct-2012
Articles in Peer-reviewed Journals	Zhang Y, Uhlemeyer J, Hada M, Asaithamby A, Chen DJ, Wu H. "Proximity within interphase chromosome contributes to the breakpoint distribution in radiation-induced kintrachromosomal exchanges." Life Sciences in Space Research. 2014 Jul;2:23-8. http://dx.doi.org/10.1016/j.lssr.2014.05.004 , Jul-2014